RAGHAVENDRA R

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https://ragha2030.github.io/ragha-portfolio/

SUMMARY

AI/ML Engineer with experience building ML models, LLM-powered applications, and RAG pipelines deployed on AWS and GCP. Skilled in end-to-end AI/ML system design, model training, evaluation, and CI/CD for scalable production.

TECHNICAL SKILLS

- Programming: Python, C/C++, SQL,
- ML &Data: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, PySpark, Matplotlib
- GenAI & NLP: LangChain, LangGraph, Prompt Engineering
- Designing: Blender

WORK EXPERIENCE

Quality Analyst **Aug 2024 - Mar 2026**

Workline Chennai, India

- Performed end-to-end Functional, Regression, UAT, API, and Integration testing to ensure high-quality software delivery aligned with business requirements.
- Managed defect tracking, root cause analysis, and cross-browser/device testing to improve application stability and user experience.
- Collaborated with Agile teams to enhance test coverage, support automation initiatives, and drive efficient, high-quality releases.

Machine Learning Intern **May 2023– Aug. 2023**

SpaceZee Technologies Chennai, India

- Built an AI assistant integrating DL based speech recognition and gesture controls improving accessibility by 30%.
- Developed a CV pipeline for real-time traffic monitoring improving accuracy by 20% and reducing latency by 15%.

EDUCATION

Oct 2020 - May 2024

SRM University

- B.Tech in Computer Science & Engineering with specialisation in Artificial Intelligence and Machine Learning (AI & ML) - CGPA: 9.03

National Public School **June 2018- May 2020**

Higher Secondary School - 80%

PROJECTS

Eye-State Classification from EEG Signals
Machine Learning

- Developed an EEG-based eye state detection model for driver assistance and healthcare applications.
- Applied and evaluated AdaBoost and Random Forest algorithms for real-time classification.
- Improved prediction accuracy by 16% using Random Forest, enhancing system performance.

Hate Speech Detection from Voice and Text using BERT
Natural Language Processing

- Built a BERT-based solution to identify hate speech from voice and text data.
- Implemented voice recognition and optimized data preprocessing for real-time performance.
- Improved detection accuracy by 14% over conventional machine learning approaches.

Traffic Management for Autonomous Vehicles Using Neural Network
Deep Learning

- Designed and implemented a neural network-based traffic simulation system for autonomous vehicles.
- Applied multi-objective and hybrid optimization techniques to enhance routing efficiency.
- Improved overall traffic performance through comprehensive evaluation and strategy optimization.

ACHIVEMENTS

Automated Aeroponics System

- ·Proposed an AI-driven Automated Aeroponics Farming System to optimize plant growth without soil, using minimal water. This system automates hydration, nutrient distribution, and lighting, reducing labor and resource needs while enabling scalable, vertical farming. Presented at SRM University Research Day and won Gold Medal for outstanding Research Contribution, Feb 2023

Emergency Alert and Adaptive Traffic Signal System for Emergency Vehicles using Machine Learning

- Expounds about Adaptive Traffic Signal System (ATSS) using AI and ML to optimize emergency vehicle response times. This system dynamically adjusts signals in real-time through object detection, RFID, and GPS inputs, reducing delays and enhancing safety. Presented at SMARTCOM 2024 , Published in Springer Journal.

CERTIFICATIONS

- Selenium Webdriver with PYTHON + Frameworks
- Microsoft Azure